Assignment 4 TCP-UPD

จัดทำโดย นายปกรณ์เกียรติ ศรีจันทร์ 673380045-2 Sec2

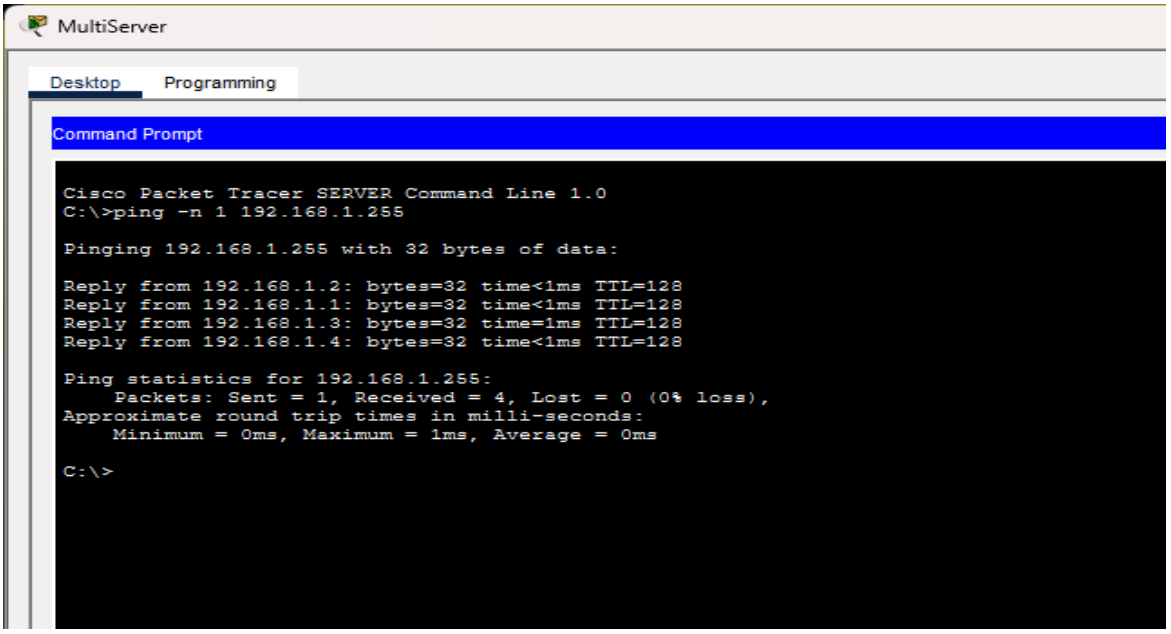
Part 1: Generate Network Traffic in Simulation Mode and View Multiplexing

Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

a. Click MultiServer and click the Desktop tab > Command Prompt.

b. Enter the ping -n 1 192.168.1.255 command. You are pinging the broadcast address for the client LAN. The command option will send only one ping request rather than the usual four. This will take a few seconds as every device on the network responds to the ping request from MultiServer.

c. Close the MultiServer window.



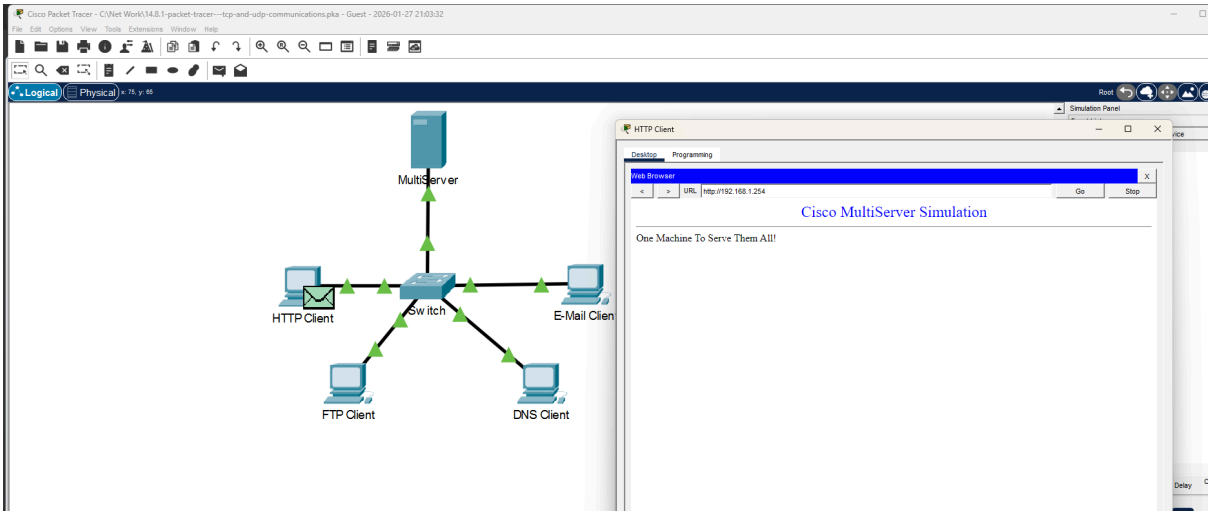
Step 2: Generate web (HTTP) traffic.

a. Switch to Simulation mode.

b. Click HTTP Client and open the Web Browser from the desktop.

c. In the URL field, enter 192.168.1.254 and click Go. Envelopes (PDUs) will appear in the topology window.

d. Minimize, but do not close, the HTTP Client configuration window.

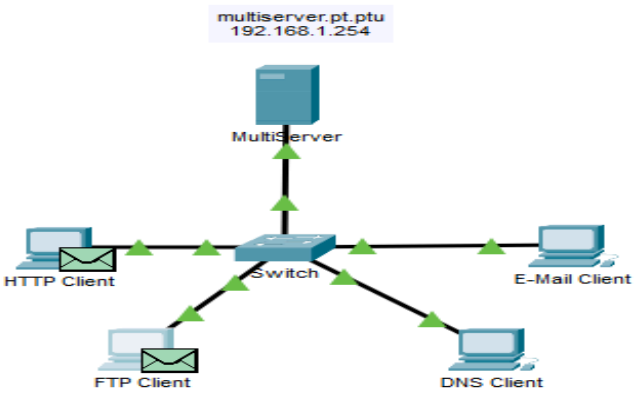


Step 3: Generate FTP traffic.

a. Click FTP Client and open the Command Prompt from the desktop

b. Enter the ftp 192.168.1.254 command. PDUs will appear in the simulation window.

c. Minimize, but do not close, the FTP Client configuration window.

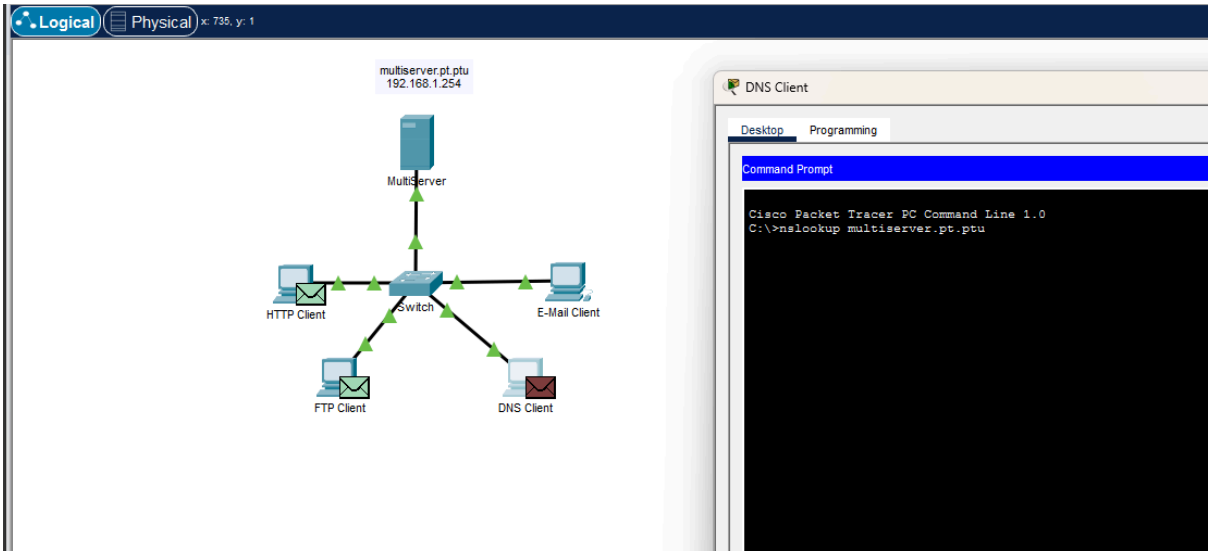


Step 4: Generate DNS traffic.

a. Click DNS Client and open the Command Prompt.

b. Enter the nslookup multiserver.pt.ptu command. A PDU will appear in the simulation window.

c. Minimize, but do not close, the DNS Client configuration window.



Step 5: Generate Email traffic.

a. Click E-Mail Client and open the E Mail tool from the Desktop.

b. Click Compose and enter the following information:

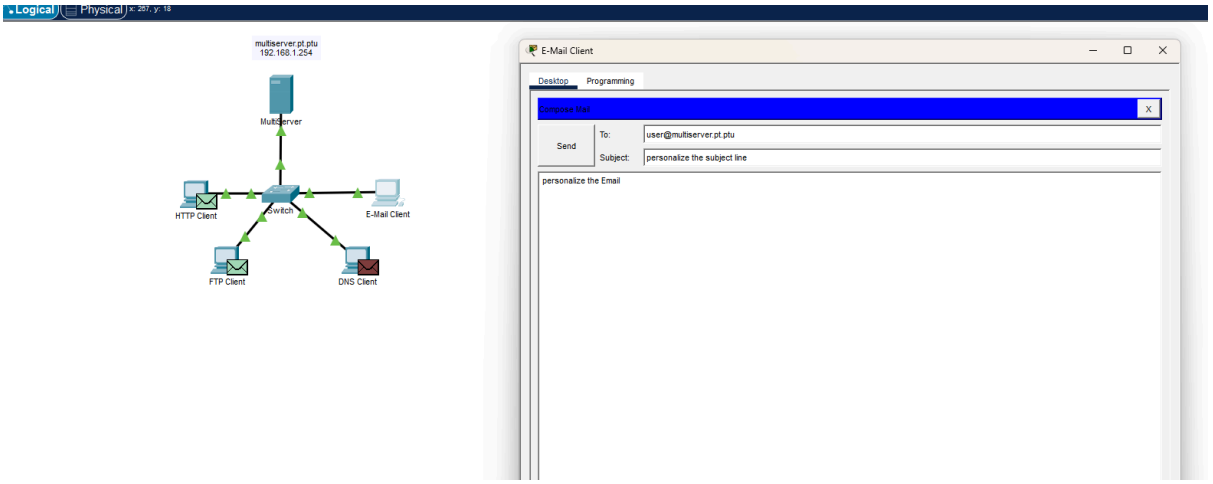
1) To: user@multiserver.pt.ptu

2) Subject: personalize the subject line

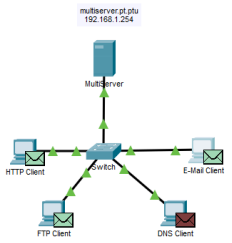
3) E-Mail Body: personalize the Email

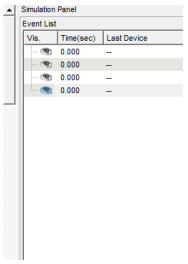
c. Click Send.

d. Minimize, but do not close, the E-Mail Client configuration window.



Step 6: Verify that the traffic is generated and ready for simulation.





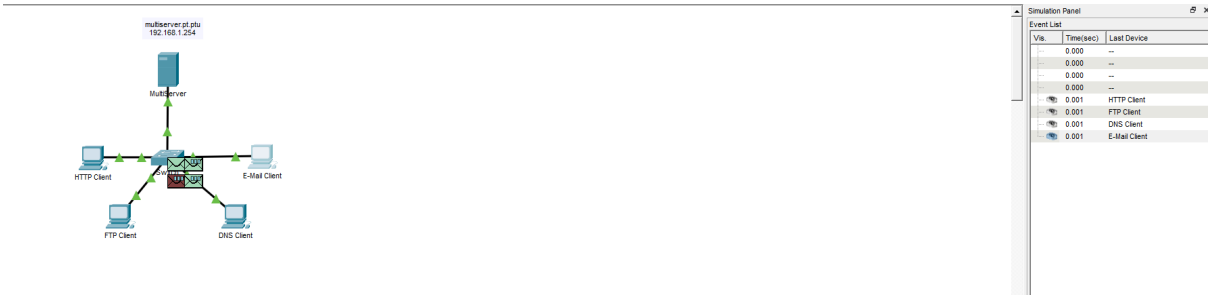
Step 7: Examine multiplexing as the traffic crosses the network.

You will now use the Capture/Forward button in the Simulation Panel to observe the different protocols travelling on the network.

Note: The Capture/Forward button ‘ >| ‘ is a small arrow pointing to the right with a vertical bar next to it.

a. Click Capture/Forward once. All of the PDUs travel to the switch.

b. Click Capture/Forward six times and watch the PDUs from the different hosts as they travel on the network. Note that only one PDU can cross a wire in each direction at any given time.



Questions:

What is this called?

Ans เรียกว่า Multiplexing

A variety of PDUs appears in the event list in the Simulation Panel. What is the meaning of the different colors?

Ans สีที่แตกต่างกันหมายถึงประเภทของโปรโตคอลที่แตกต่างกัน

Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine HTTP traffic as the clients communicate with the server.

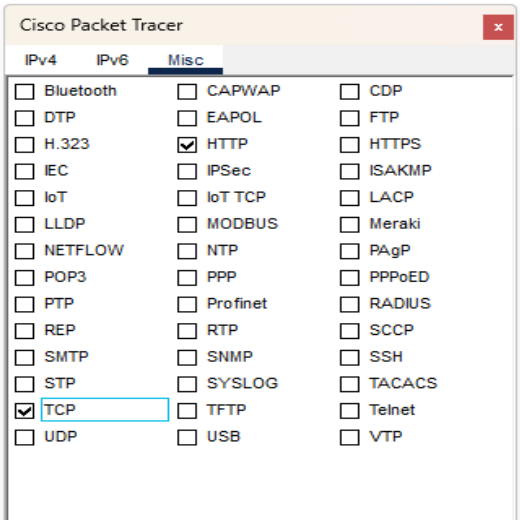
a. Click Reset Simulation.

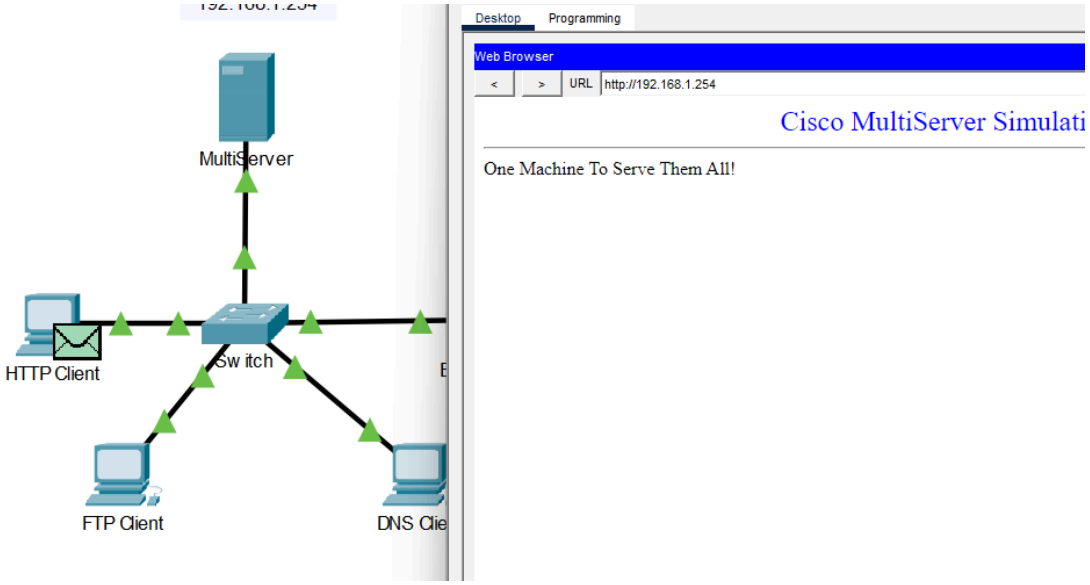
b. Filter the traffic that is currently displayed to only HTTP and TCP PDUs. To filter the traffic that is currently displayed:

1) Click Edit Filters and toggle the Show All/None button.

2) Select HTTP and TCP. Click the red “x” in the upper right-hand corner of the Edit Filters box to close it. Visible Events should now display only HTTP and TCP PDUs.

c. Open the browser on HTTP Client and enter 192.168.1.254 in the URL field. Click Go to connect to the server over HTTP. Minimize the HTTP Client window.





d. Click Capture/Forward until you see a PDU appear for HTTP. Note that the color of the envelope in the topology window matches the color code for the HTTP PDU in the Simulation Panel.

Why did it take so long for the HTTP PDU to appear?

Ans ต้องมีการทำ TCP Three-Way Handshake เพื่อสร้างการเชื่อมต่อก่อนที่ HTTP จะเริ่มส่งข้อมูลได้

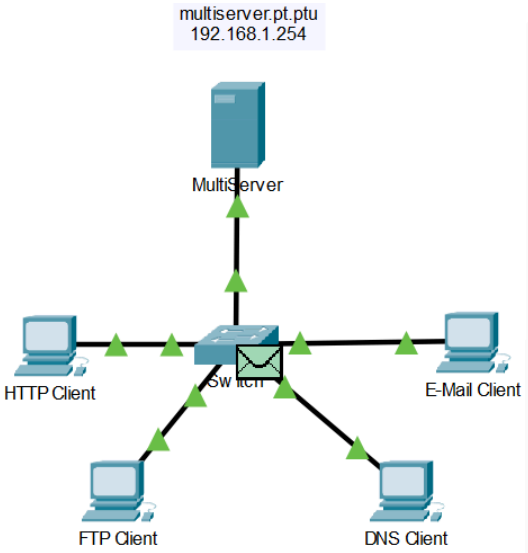
e. Click the PDU envelope to show the PDU details. Click the Outbound PDU Details tab and scroll down to the second to the last section.

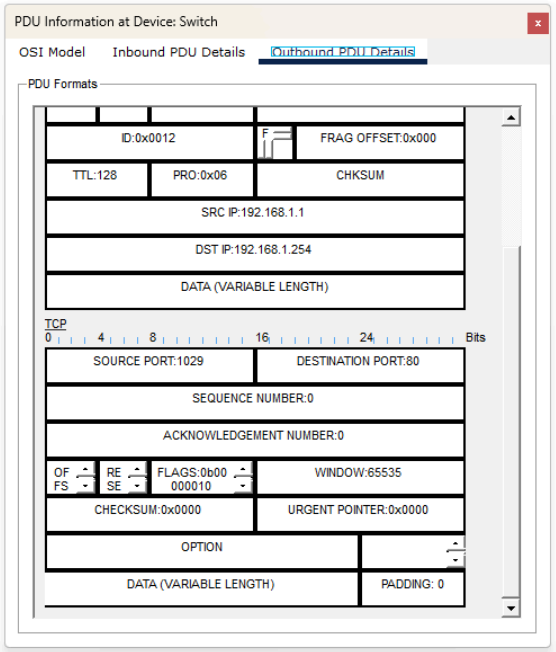
What is the section labeled?

Ans TCP

Are these communications considered to be reliable?

Ans ใช่ เพราะ TCP มีกลไก Acknowledgment, การจัดลำดับข้อมูล และการส่งใหม่เมื่อข้อมูลสูญหาย เพื่อรับประกันความถูกต้อง





Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values.

Ans SRC PORT=1029

DEST PORT=80

SEQUENCE NUM=1

ACK NUM =1

f.Look at the value in the Flags field, which is located next to the Window field. The values to the right of the “b” represent the TCP flags that are set for this stage of the data conversation. Each of the six places corresponds to a flag. The presence of a “1” in any place indicates that the flag is set. More than one flag can be set at a time. The values for the flags are shown below.

Flag Place	6	5	4	3	2	1
Value	URG	ACK	PSH	RST	SYN	FIN

Flags: 0b00000010

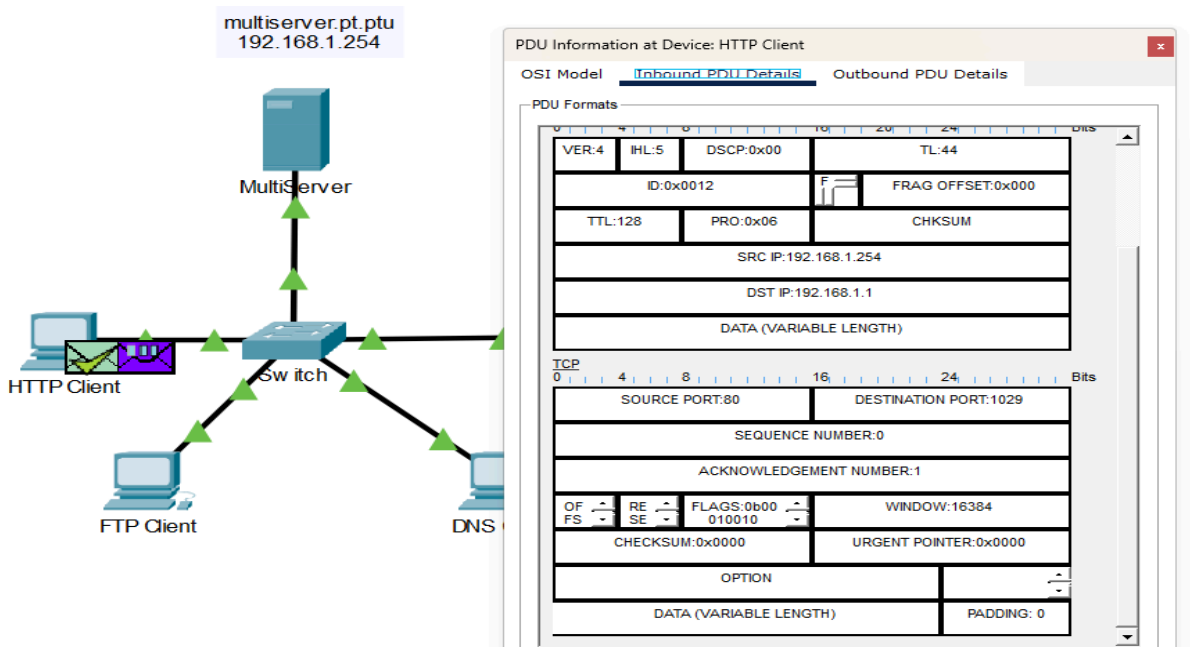
ลำดับ (Flag Place)	6	5	4	3	2	1
ชื่อ Flag (Value)	URG	ACK	PSH	RST	SYN	FIN
ค่าที่อ่านได้	0	0	0	0	1	0

Which TCP flags are set in this PDU?

Ans SYN

g. Close the PDU and click Capture/Forward until a PDU with a checkmark returns to the HTTP Client.

h. Click the PDU envelope and select Inbound PDU Details.



Question:

How are the port and sequence numbers different than before?

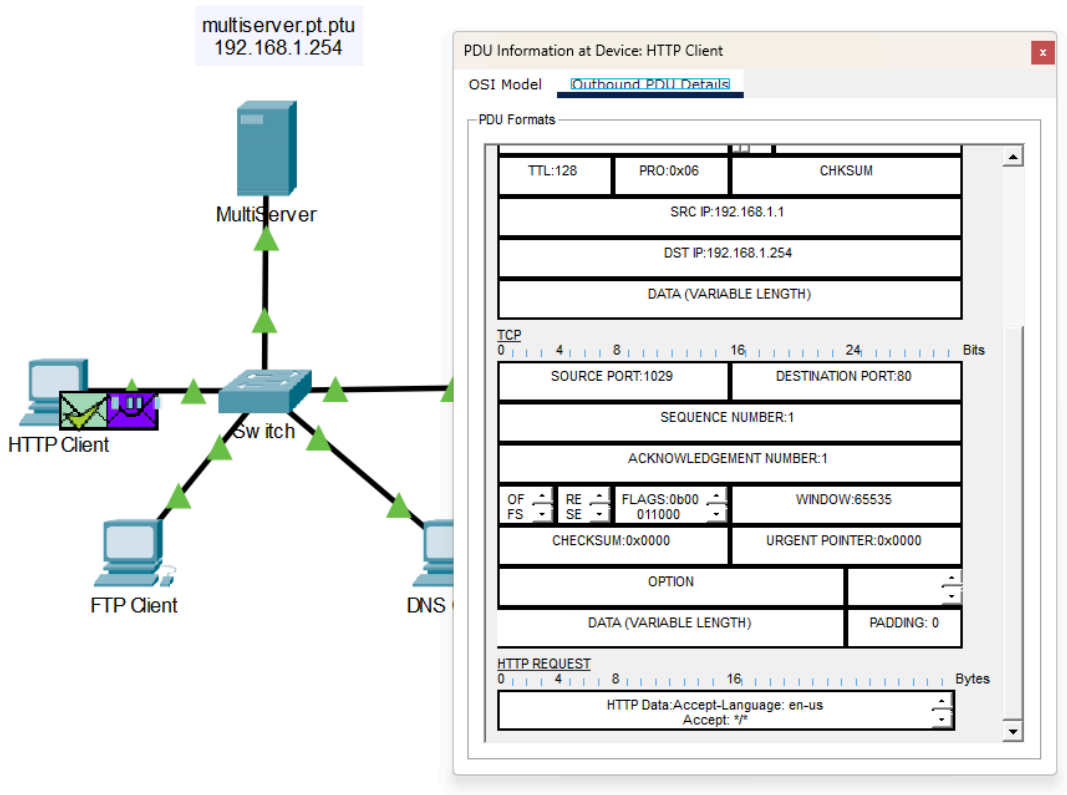
Ans SRC PORT = 80

DEST PORT = 1029

SEQUENCE NUM = 0

ACK NUM = 1

i. Click the HTTP PDU which HTTP Client has prepared to send to MultiServer. This is the beginning of the HTTP communication. Click this second PDU envelope and select Outbound PDU Details.



Question:

What information is now listed in the TCP section? How are the port and sequence numbers different

from the previous two PDUs?

Ans

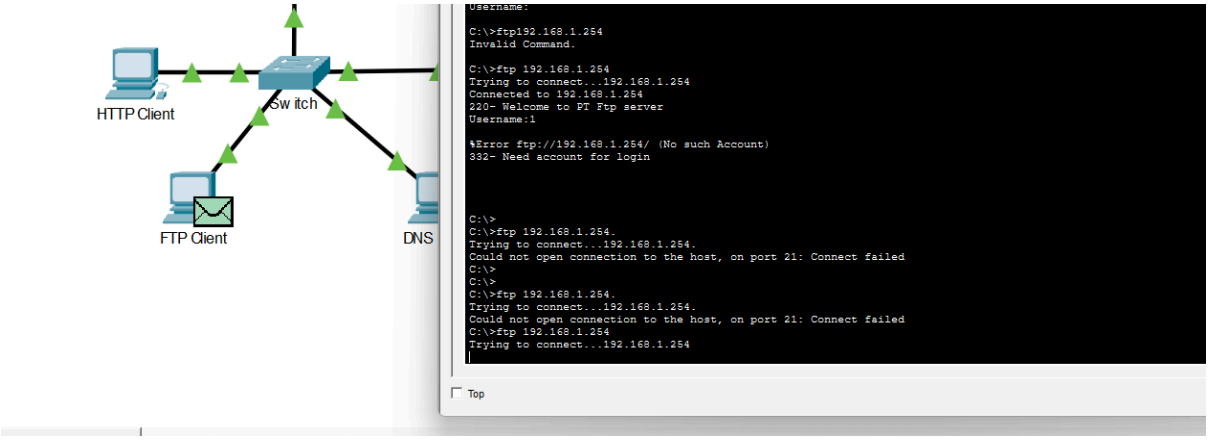
รายการ	PDU 1 (SYN)	PDU 2 (SYN/ACK)	PDU ปัจจุบัน (HTTP Req)
Source Port	1029	80	1029
Dest Port	80	1029	80
Sequence Number	0	0	1
Ack Number	0	1	1

j. Reset the simulation.

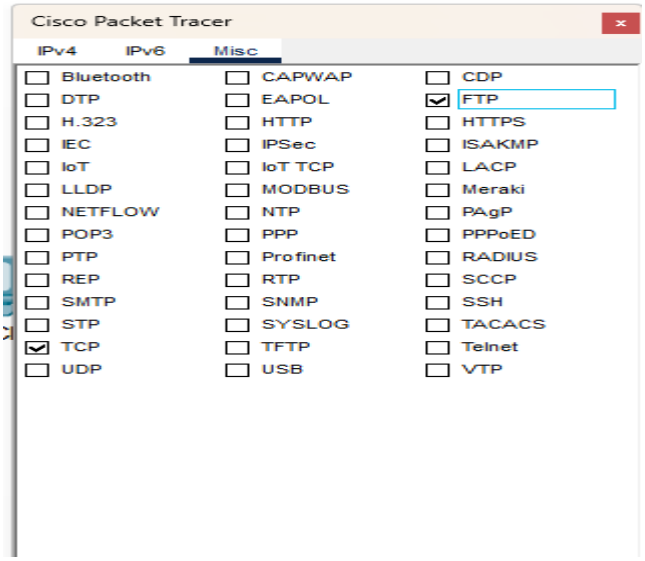
Step 2: Examine FTP traffic as the clients communicate with the server.

a. Open the command prompt on the FTP Client desktop. Initiate an FTP connection by entering ftp

192.168.1.254.

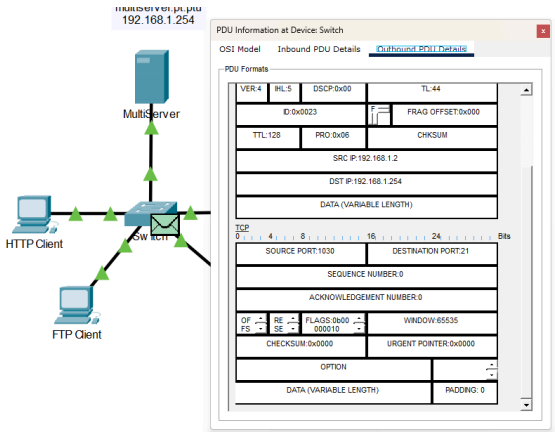


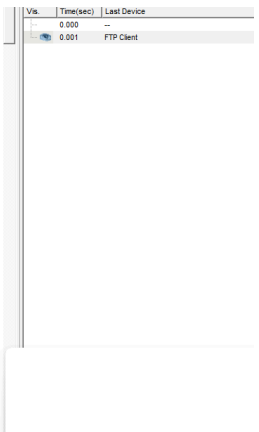
b. In the Simulation Panel, change Edit Filters to display only FTP and TCP.



c. Click Capture/Forward. Click the second PDU envelope to open it.

Click the Outbound PDU Details tab and scroll down to the TCP section.





Question:

Are these communications considered to be reliable?

Ans การสื่อสารนี้ถือว่ามีความน่าเชื่อถือ

d. Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values.

Ans SRC PORT : 1030

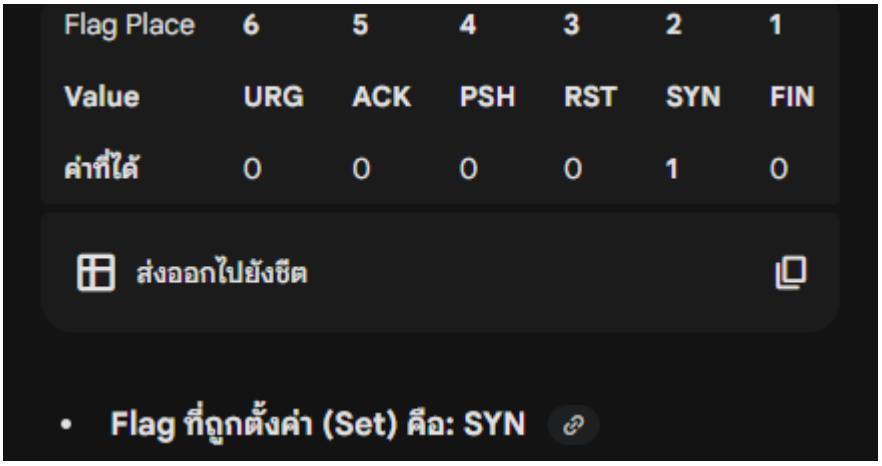
DEST PORT : 21

SEQUENCE NUM: 0

ACK NUM: 0

What is the value in the flag field?

Ans Flag Field Value: 0b00000010

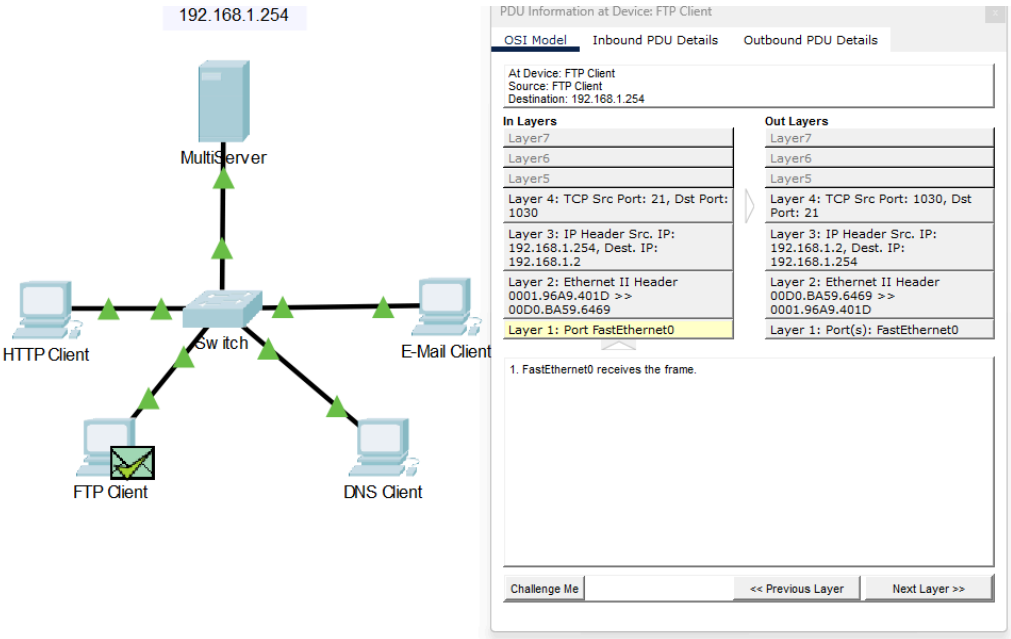


Flag Place	6	5	4	3	2	1
Value	URG	ACK	PSH	RST	SYN	FIN
ค่าที่ได้	0	0	0	0	1	0

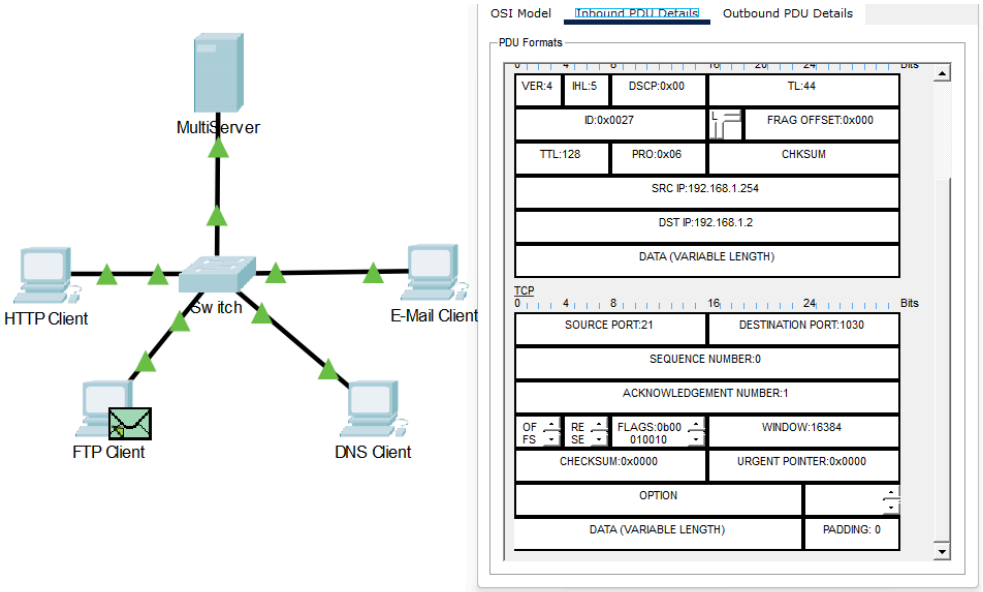
ส่งออกไปยังชีต

- Flag ที่ถูกตั้งค่า (Set) คือ: SYN

e. Close the PDU and click Capture/Forward until a PDU returns to the FTP Client with a checkmark.



f. Click the PDU envelope and select Inbound PDU Details.



Question:

How are the port and sequence numbers different than before?

Ans SRC PORT : 21

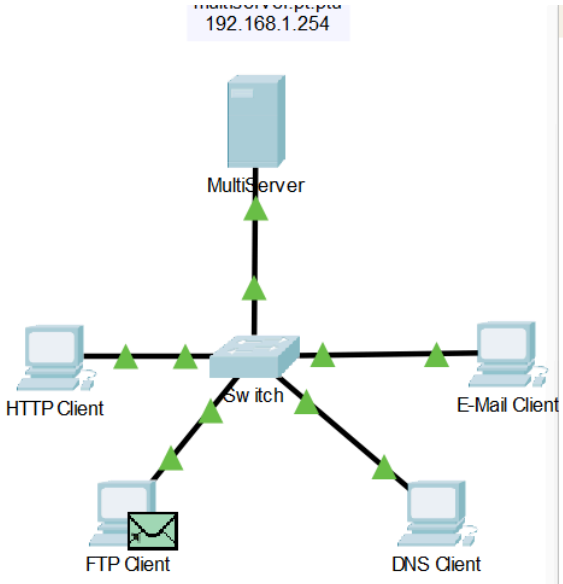
DEST PORT : 1030

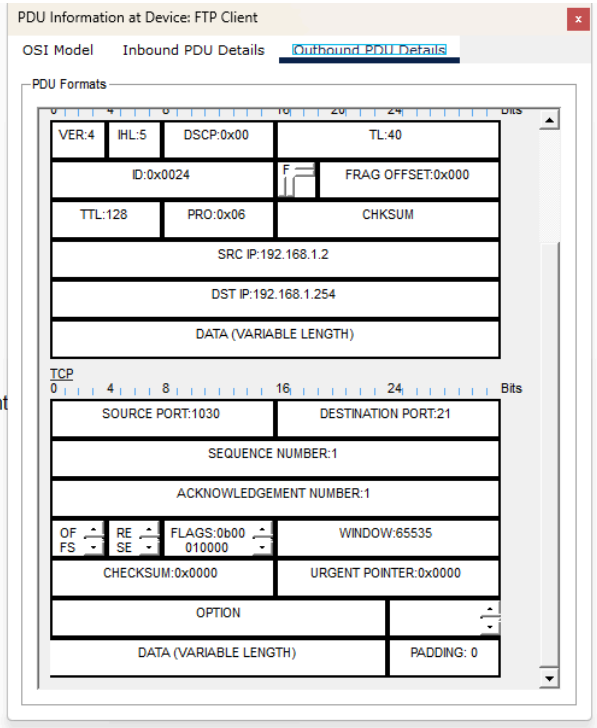
SEQUENCE NUM: 0

ACK NUM: 1

หมายเลขพอร์ตต้นทางและปลายทางมีการสลับตำแหน่งกันเนื่องจากเป็นการส่งข้อมูลย้อนกลับ และค่า Acknowledgement Number เพิ่มขึ้นจาก 0 เป็น 1 เพื่อยืนยันว่าได้รับแพ็กเกจแรกเรียบร้อยแล้ว

g. Click the Outbound PDU Details tab.



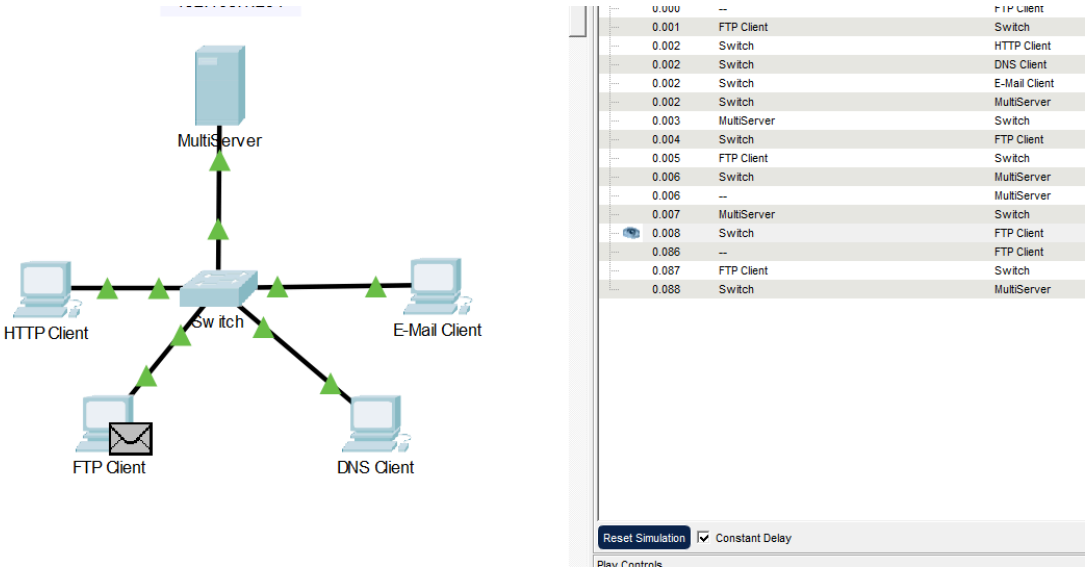


Question:

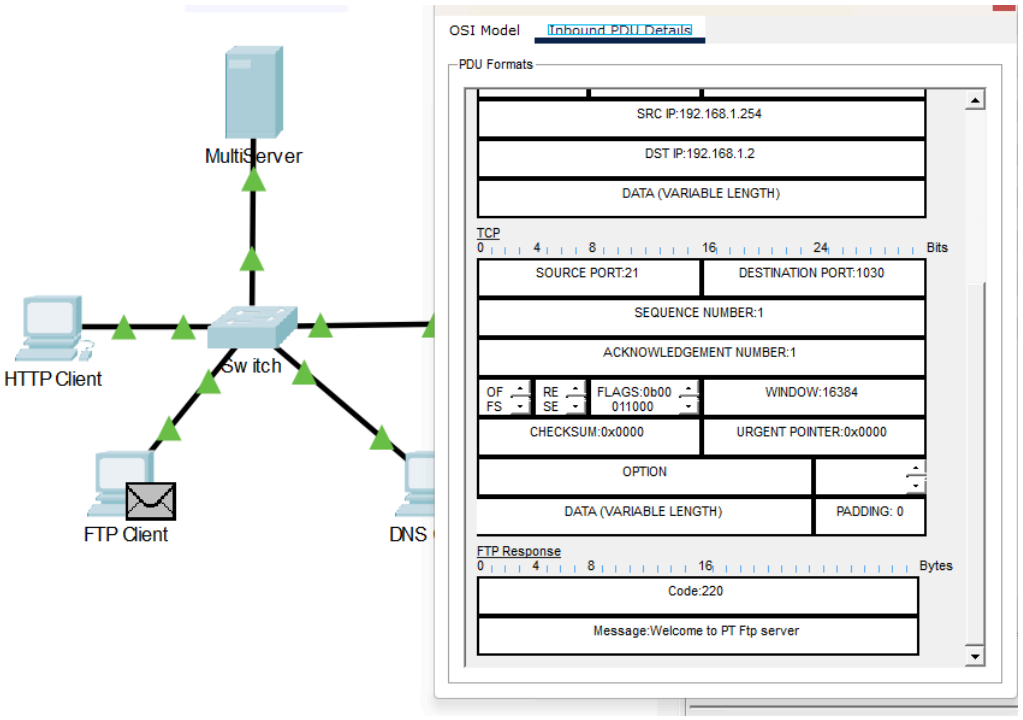
How are the port and sequence numbers different from the previous results?

Ans Port Numbers ยังคงเป็นคู่เดิมคือ Source: 1030 และ Destination: 21 ส่วน Sequence และ Acknowledgement Numbers เพิ่มขึ้นเป็น 1 เนื่องจากสิ้นสุดขั้นตอน Handshake และเข้าสู่สถานะ Established เพื่อเริ่มรับส่งคำสั่งของโปรโตคอล FTP

h. Close the PDU and click Capture/Forward until a second PDU returns to the FTP Client. The PDU is a different color.



i. Open the PDU and select Inbound PDU Details. Scroll down past the TCP section.



Question:

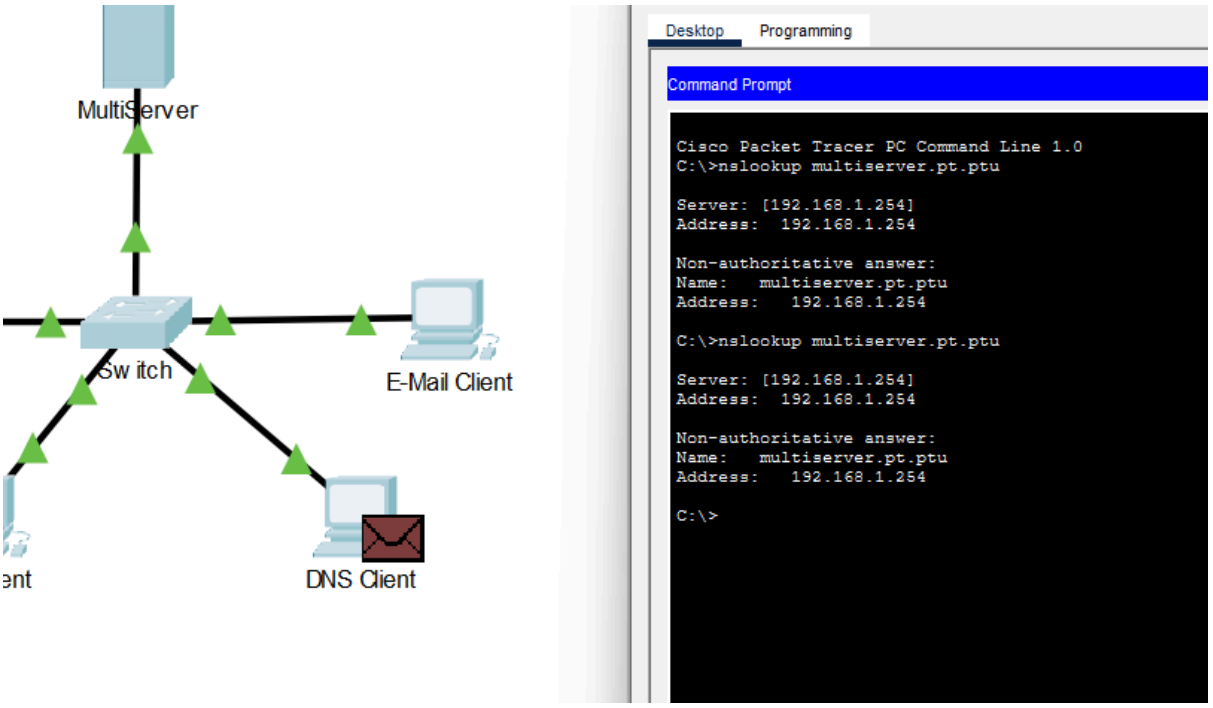
What is the message from the server?

Ans Message:Welcome to PT Ftp server

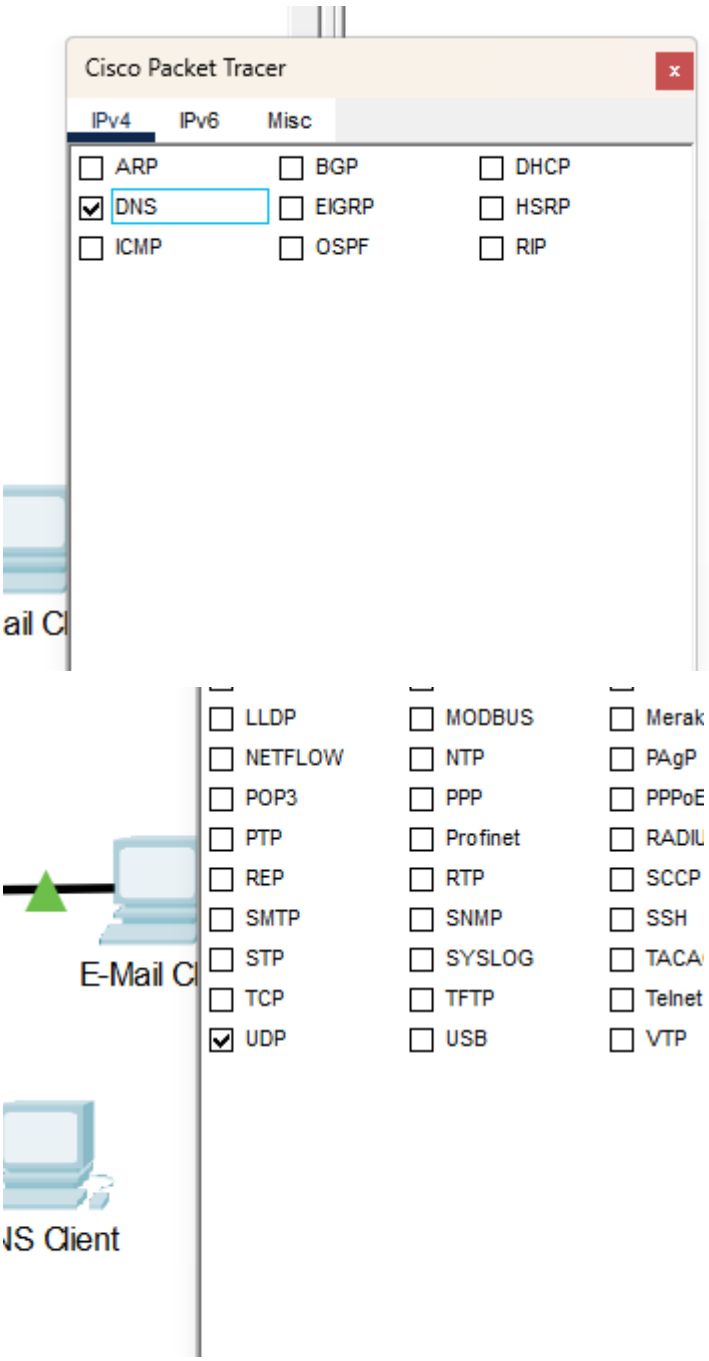
j. Click Reset Simulation.

Step 3: Examine DNS traffic as the clients communicate with the server.

a. Repeat the steps in Part 1 to create DNS traffic.



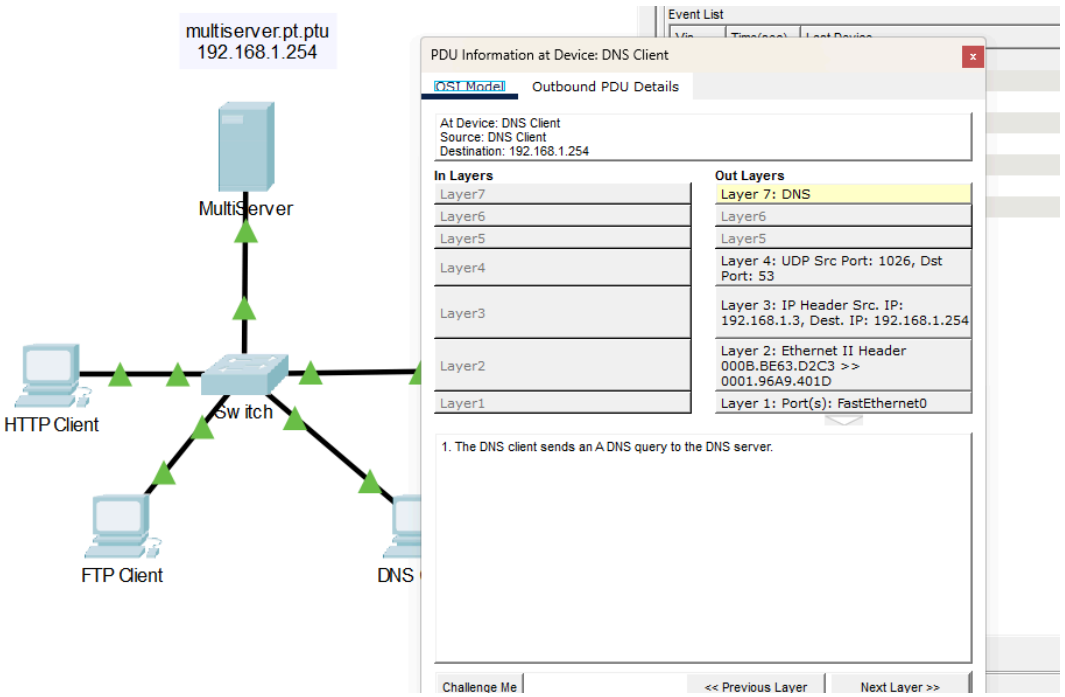
b. In the Simulation Panel, change Edit Filters to display only DNS and UDP.



c. Click the PDU envelope to open it.

d. Look at the OSI Model details for the outbound PDU.

Question:



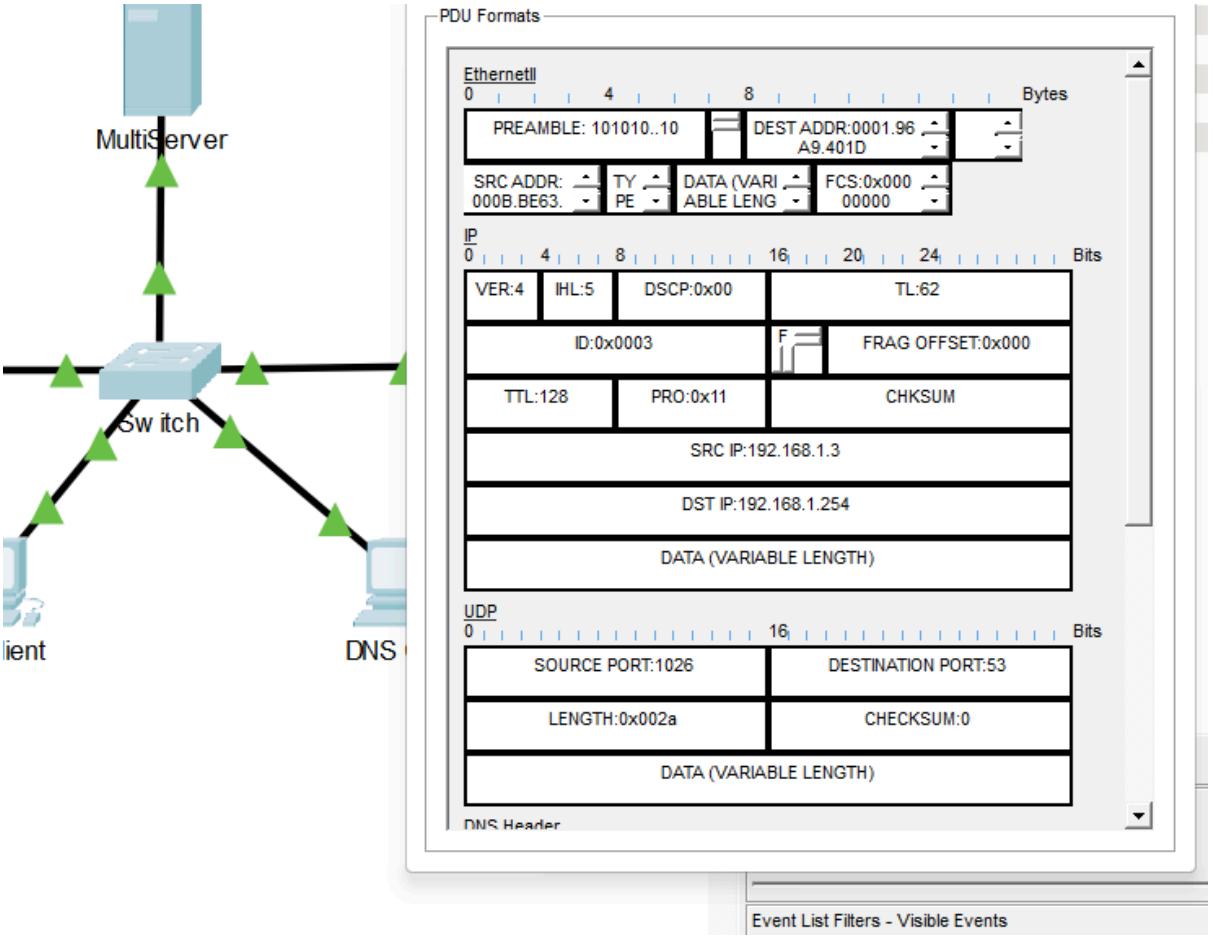
What is the Layer 4 protocol?

Ans UDP

Are these communications considered to be reliable?

Ans การสื่อสารในส่วนของ DNS นี้ไม่ถือว่าเป็นการสื่อสารที่เชื่อถือได้ (Not considered to be reliable) ในเชิงของโปรโตคอลระดับ Transport Layer

e. Open the Outbound PDU Details tab and find the UDP section of the PDU formats. Record the SRC PORT and DEST PORT values.



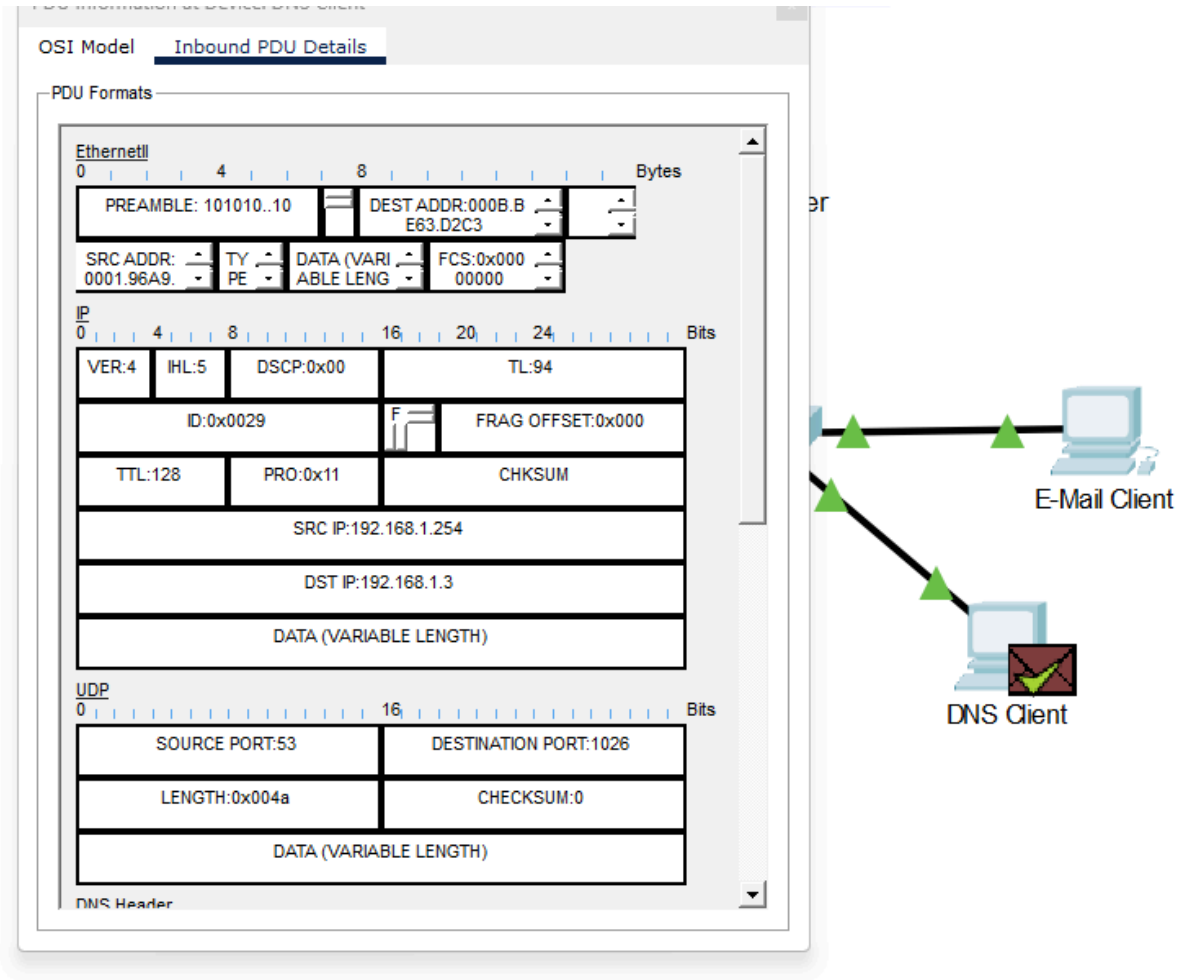
Question:

Why are there no sequence and acknowledgement numbers?

Ans เพราะ DNS ทำงานบนโปรโตคอล UDP ซึ่งเป็นแบบ Connectionless จึงไม่มีฟิลด์ Sequence และ Acknowledgement เพื่อลด Overhead และเน้นความรวดเร็วในการรับส่งข้อมูล

f. Close the PDU and click Capture/Forward until a PDU with a check mark returns to the DNS Client.

g. Click the PDU envelope and select Inbound PDU Details.



Question:

How are the port and sequence numbers different than before?

Ans Outbound

Source Port = 1026

Destination Port = 53

Inboun

Source Port = 53

Destination Port = 1026

What is the last section of the PDU called? What is the IP address for the name multiserver.pt.ptu?

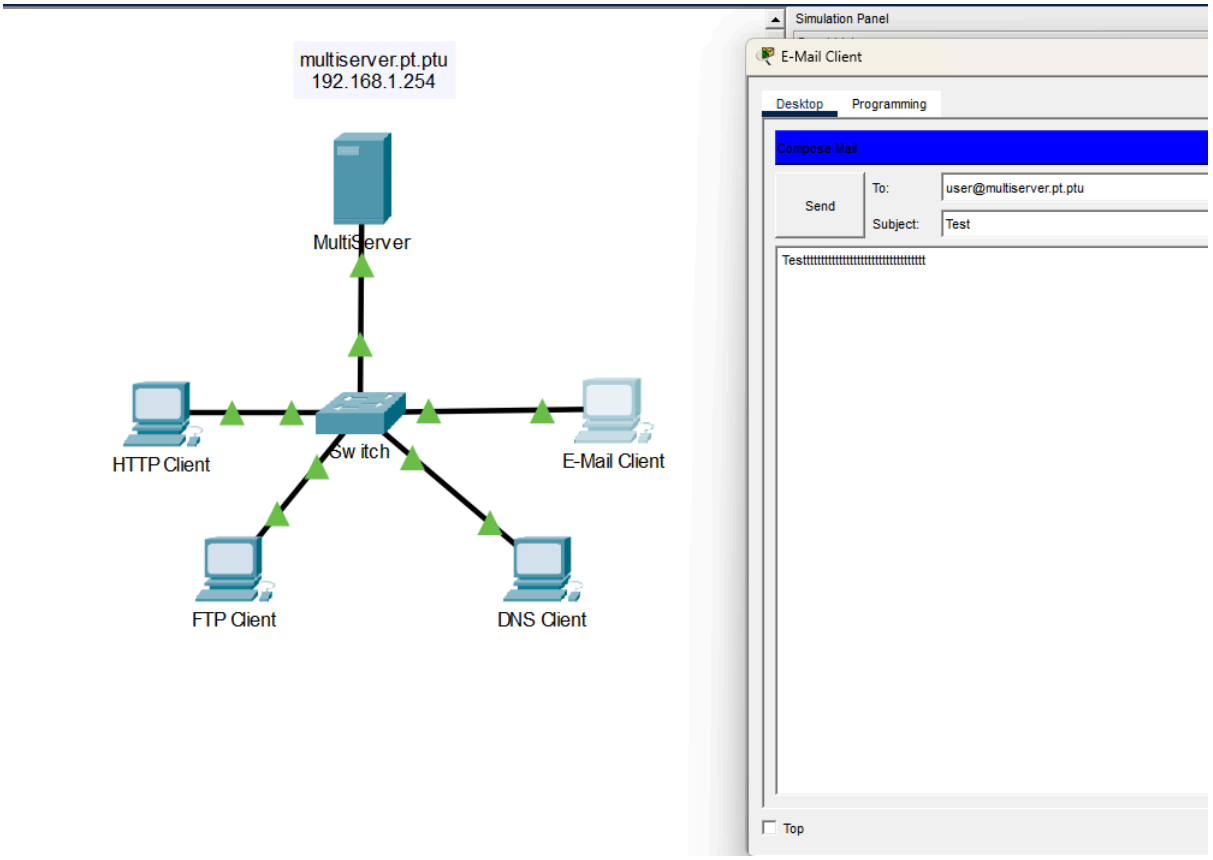
Ans เมื่อพิจารณาจาก PDU ขากลับ ส่วนสุดท้ายจะเรียกว่า DNS Answer

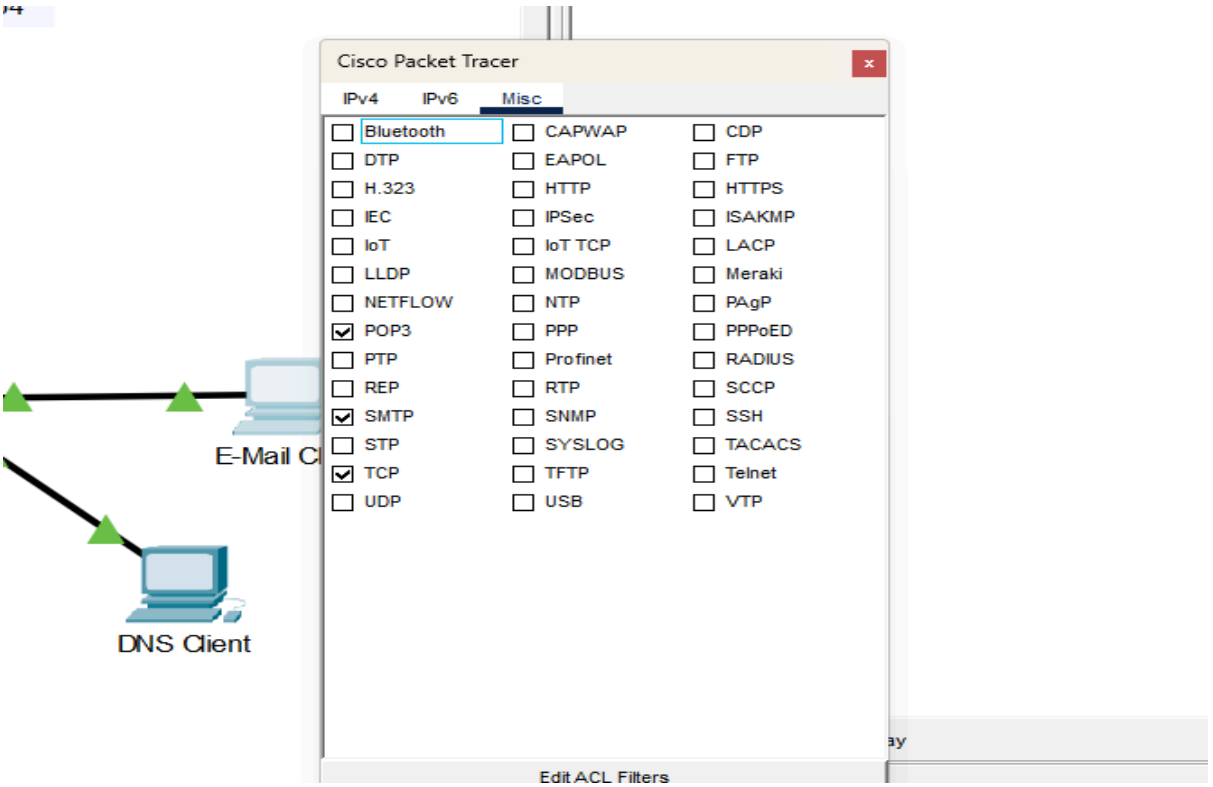
and multiserver.pt.ptu 192.168.1.254

h. Click Reset Simulation.

Step 4: Examine email traffic as the clients communicate with the server.

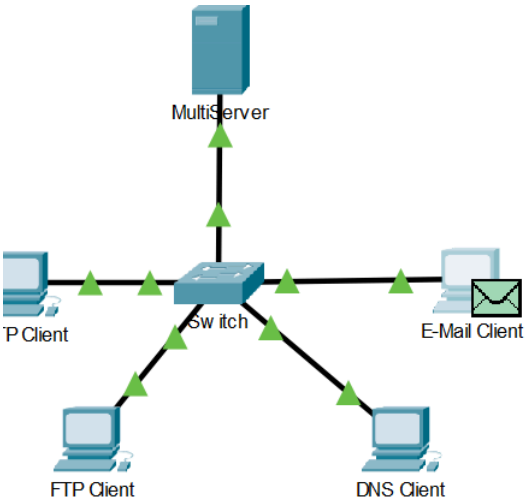
a. Repeat the steps in Part 1 to send an email to user@multiserver.pt.ptu.

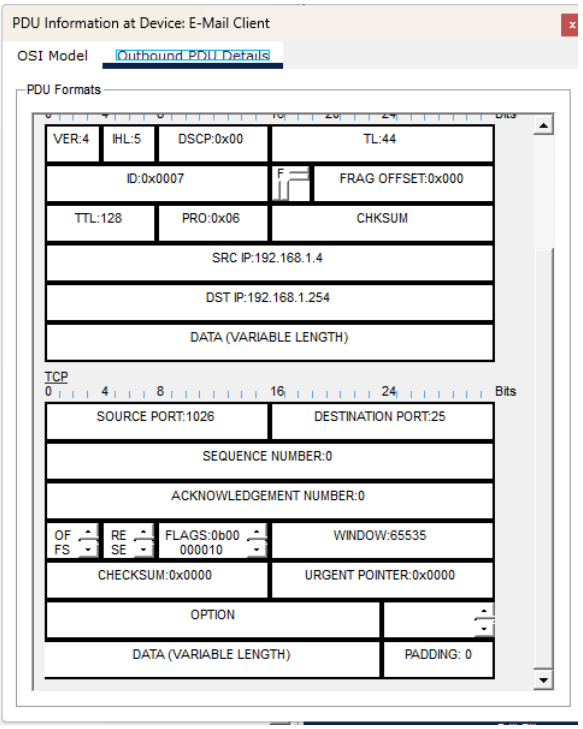

multiserver.pt.ptu
192.168.1.254
MultiServer
HTTP Client
Switch
E-Mail Client
FTP Client
DNS Client
Simulation Panel
E-Mail Client
Desktop
Programming
Compose Mail
Send
To:
user@multiserver.pt.ptu
Subject:
Test
Testtttttt



c. Click the first PDU envelope to open it.

d. Click the Outbound PDU Details tab and scroll down to the last





section.

Questions:

What transport layer protocol does email traffic use?

Ans TCP

Are these communications considered to be reliable?

Ans เชื่อถือได้ เพราะ TCP มีการรับประกันการส่งข้อมูล

e. Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM

SRC PORT :1026

DEST PORT : 25

SEQUENCE NUM: 0

ACK NUM: 0

values. What is the flag field

value?

Ans Flag Field Value: 0b00000010

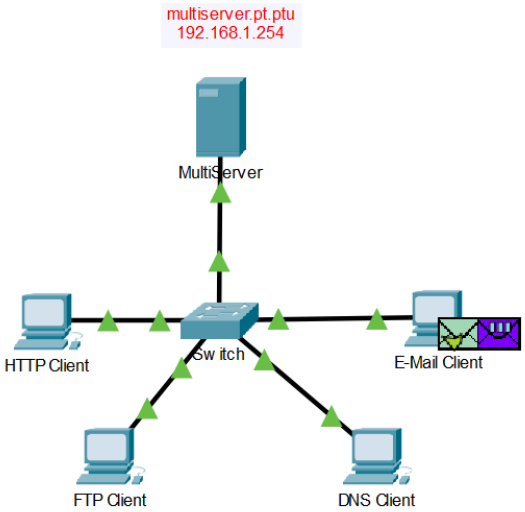
Flag Place (ตำแหน่ง)	6	5	4	3	2	1
ชื่อ Flag (Value)	URG	ACK	PSH	RST	SYN	FIN
ค่าที่ได้ (000010)	0	0	0	0	1	0

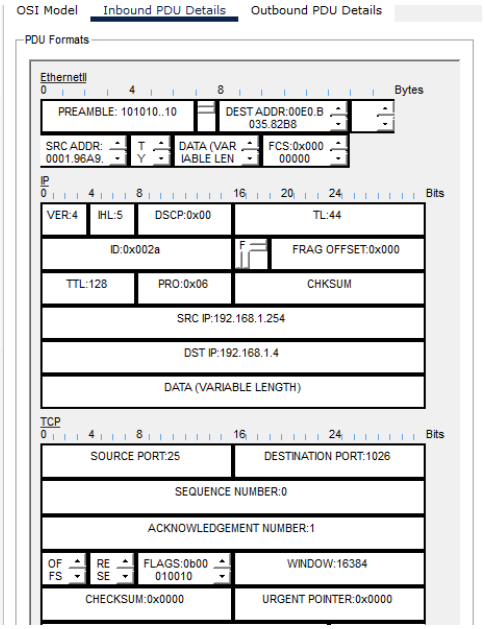
Flag ที่ถูกตั้งค่าคือ: SYN

ความหมาย: ค่านี้มักพบในโปรโตคอล TCP (เช่น HTTP) ในขั้นตอนแรกของการเริ่มขอเชื่อมต่อ Three-way Handshake เพื่อตกลงหมายเลขลำดับเริ่มต้น

f. Close the PDU and click Capture/Forward until a PDU returns to the E-Mail Client with a checkmark.

g. Click the TCP PDU envelope and select Inbound PDU Details.





Question:

How are the port and sequence numbers different than before?

ANS SRC PORT :25

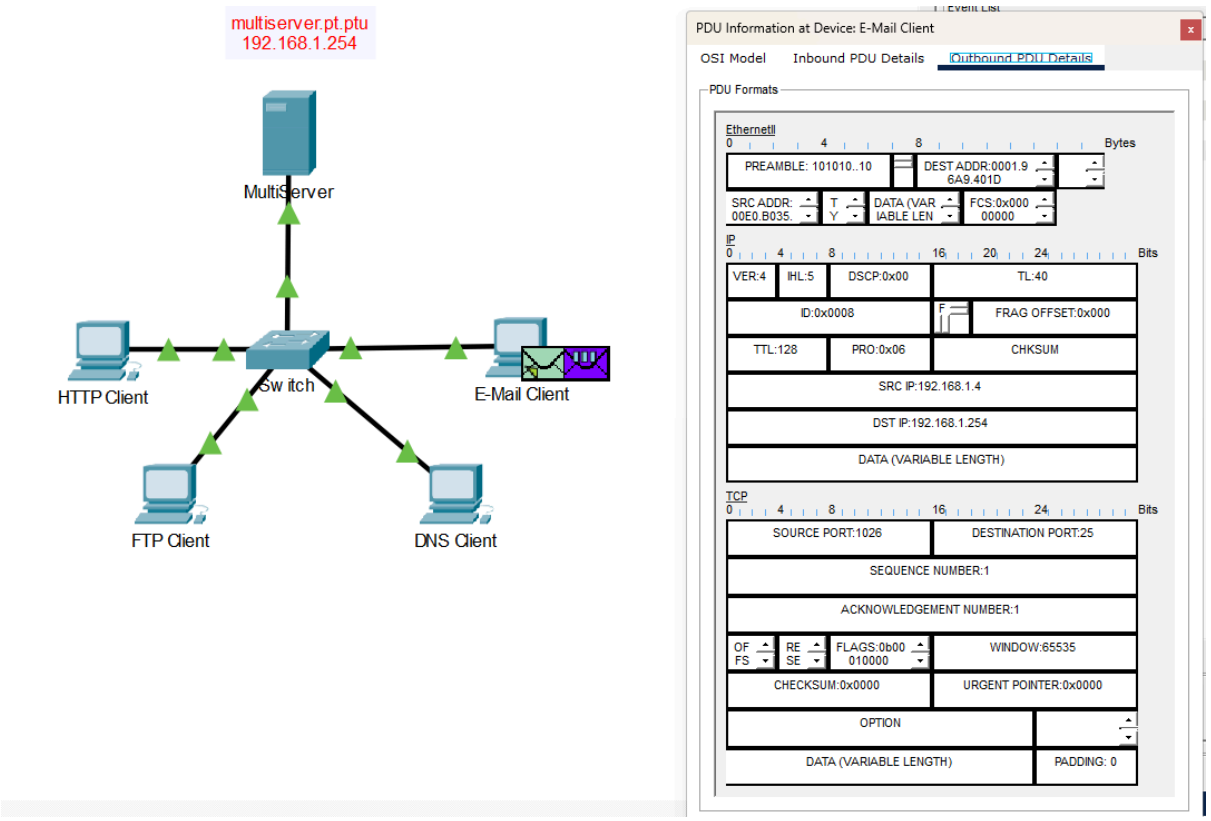
DEST PORT : 1026

SEQUENCE NUM: 0

ACK NUM: 1

หมายเลขพอร์ตต้นทางและปลายทางมีการสลับตำแหน่งกันเนื่องจากเป็นการส่งข้อมูลย้อนกลับจากเซิร์ฟเวอร์ และค่า Acknowledgement Number เพิ่มขึ้นจาก 0 เป็น 1 เพื่อยืนยันว่าได้รับแพ็กเกจเชื่อมต่อแรกเรียบร้อยแล้ว

Click the Outbound PDU Details tab.



Question:

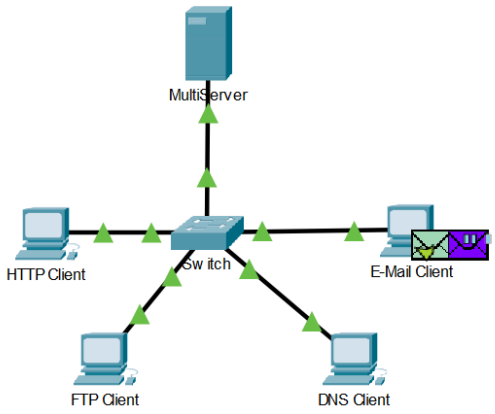
How are the port and sequence numbers different from the previous two results?

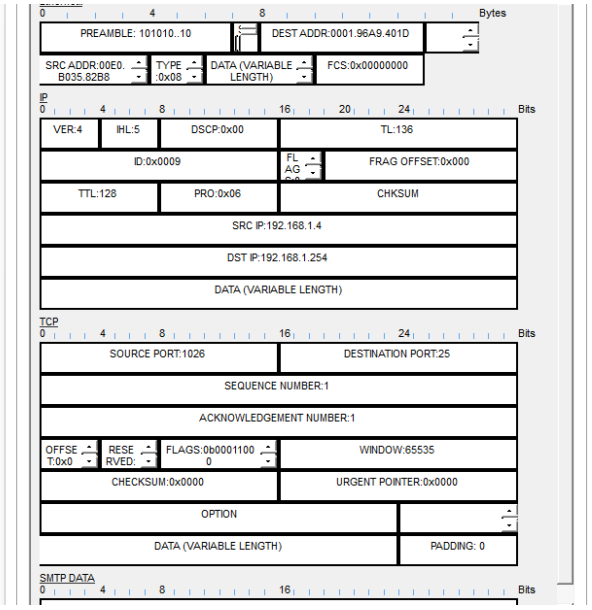
ANS Port Numbers กลับมาใช้คู่เดิมคือ Source: 1026 และ Destination: 25 ส่วน Sequence และ Acknowledgement Numbers เปลี่ยนจาก 0 เป็น 1 เนื่องจากกระบวนการ Handshake เสร็จสิ้นและเข้าสู่สถานะเริ่มรับส่งข้อมูล

There is a second PDU of a different color that E-Mail Client has prepared to send to MultiServer. This

is the beginning of the email communication. Click this second PDU envelope and select Outbound PDU

Details.





Questions:

How are the port and sequence numbers different from the previous two PDUs?

ANS Port Numbers: กลับมาใช้ Source: 1026 และ Destination: 25 (ทิศทาง Client ไปยัง Server)

Seq/Ack Numbers: มีค่าเป็น 1 ทั้งคู่

คำอธิบาย: เนื่องจากกระบวนการ Three-way Handshake เสร็จสิ้นแล้ว ตัวเลขจึงขยับจาก 0 เป็น 1 เพื่อเริ่มต้นขั้นตอนการรับส่งข้อมูลของโปรโตคอล SMTP

What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?

Ans TCP Port 25: เกี่ยวข้องกับโปรโตคอล SMTP (Simple Mail Transfer Protocol)

- หน้าที่: ใช้สำหรับการ "ส่ง" อีเมลจากเครื่องลูกข่ายไปยังเซิร์ฟเวอร์ หรือระหว่างเซิร์ฟเวอร์ด้วยกันเอง
- หลักฐานจาก PDU: ในรายละเอียด PDU ที่ใช้พอร์ตปลายทาง (Destination Port) เป็น 25 จะปรากฏส่วนข้อมูลที่ระบุว่าเป็น SMTP DATA อยู่ที่ส่วนท้ายสุดของแพ็กเกจ

TCP Port 110: เกี่ยวข้องกับโปรโตคอล POP3 (Post Office Protocol version 3)

- หน้าที่: ใช้สำหรับการ "รับ" หรือดึงอีเมลจากเซิร์ฟเวอร์ลงมายังเครื่องลูกข่าย